



Short communication

Fast pyrolysis of empty fruit bunches

N. Abdullah^{a,*}, H. Gerhauser^b, F. Sulaiman^a^a School of Physics, Universiti Sains Malaysia, 11800 Minden, Penang, Malaysia^b Biomass, Coal and the Environment Unit, Energy Research Centre of the Netherlands (ECN), 1755 ZG Petten, The Netherlands

ARTICLE INFO

Article history:

Received 5 August 2009

Received in revised form 17 December 2009

Accepted 17 December 2009

Available online 31 December 2009

Keywords:

Empty fruit bunches

Fast pyrolysis

Oil palm

Bio-oil

Biomass

ABSTRACT

This short communication describes the evaluation of the fast pyrolysis behaviour of empty fruit bunches (EFB), one of the solid wastes of the rapidly expanding palm oil industry. A 150 g/h fluidised bed bench scale fast pyrolysis unit is used to study the impact of the following key variables; the reactor temperature in the range of 400–600 °C, the residence time in the range of 0.79–1.32 s, and a range of particle sizes (with percentage of ash content) obtained by sieving of <150 µm (8.49%), 150–250 µm (7.46%), 250–300 µm (6.70%) and 355–500 µm (4.83%). The results confirmed the shape of the yield curve for EFB and indicated the significant difference when comparing the literature values for yields with the results obtained in this study along with the systems being used.

© 2009 Elsevier Ltd. All rights reserved.

1. Introduction

Palm oil production is on a steeply rising path. According to the US Department of Agriculture, it narrowly overtook soybean oil in the agricultural year 2004–2005 for the first time and was the vegetable oil with the largest production volume [1]. This feat was achieved on roughly a tenth of the land of the soybean crop which illustrates how productive the oil palm is. It is the highest yielding crop among major oil crops [2].

A particularly interesting waste by-product are empty fruit bunches. Typically palm oil mills use shell and the drier part of the fibre wastes, rather than EFB, to fuel their boilers, as the raw EFB contain nearly 60% water [3].

Fast pyrolysis represents a potential route to upgrade the EFB waste to value added fuels and renewable chemicals. For woody feedstocks, temperatures around 500 °C together with short vapour residence times are used to obtain bio-oil yields of around 70%, along with char and gas yields of around 15% each [4]. In an earlier full paper [5] by the authors of this short communication, the fast pyrolysis of washed EFB was considered, which was similar to that of woody feedstocks, when the alkali content had been sufficiently reduced. In this short communication, further results on the unwashed feedstock are reported.

2. Experimental

Empty fruit bunches (EFB) used in the experiments were supplied by the Malaysian Palm Oil Board. Samples received in the form of whole bunches, therefore, the bunches were chopped into smaller sizes. Then, a Fritsch grinder with a screen size of 500 µm was used to reduce the size of the feedstock to less than 500 µm. The particle sizes of interest for this study are between 250 and 355 µm as the feedstock of this size range can easily be fed into the feeder.

The moisture content of the feedstock was also measured to ensure that it was less than 10 wt.% on a dry feed basis. NREL Standard Analytical Method LAP005 for ash analysis was carried out five times for each sample and the range of variation was relatively narrow [5.27–5.72 mf wt.%]. Table 1 shows that with sieved feedstock the ash content is highest for the smallest particle size fraction. It is expected that a lot of inorganic soil fine particles could be found in the smallest particle size fraction. The mass average ash content from Table 1 is 5.39 mf wt.%.

The present work was carried out on a fluidised bed reactor with a nominal capacity of 150 g/h. Fig. 1 shows the schematic diagram of the fluidised bed pyrolysis system which consists of three main parts, the feeder, reactor and product collection. The reactor consists of a 316 stainless steel cylinder with a length of 260 mm and an internal diameter of 40 mm. The heating medium in the reactor is inert sand of size between 355 and 500 µm. The sand fills the reactor to a depth of approximately 8 cm and expands during fluidization to 12 cm. The fluidising gas is nitrogen, which is pre-heated in its flow line by the tube furnace prior to entering the base of the reactor.

* Corresponding author. Tel.: +60 4 6532475; fax: +60 4 6579150.
E-mail address: nurham2299@yahoo.com (N. Abdullah).

Table 1

Ash content of untreated EFB by size fraction.

Feed particle size (μm)	Mass fraction	Average ash (mf wt.%)
Less than 250	22	7.44
250–355	30	5.29
355–500	42	4.82
More than 500	6	4.72
Mass average	–	5.39

To investigate the effect of temperature, a series of fast pyrolysis experiments were carried out at a vapour residence time of 1.02–1.05 s on feedstock of size 250–355 μm over the temperature range of 400–600 °C in 25 °C increments.

The impact of varying vapour residence time in the range of 0.79–1.32 s by changing the nitrogen flow rate was investigated in six experiments employing a fluidised bed temperature of 500 °C and a feedstock size range of 300–355 μm .

The combined effect of ash content and particle size were assessed through further four experiments in the temperature range of 500–520 °C considering the following size fractions with ash content given in brackets in moisture free weight percent: <150 μm (8.49%), 150–250 μm (7.46%), 250–300 μm (6.70%) and 355–500 μm (4.83%).

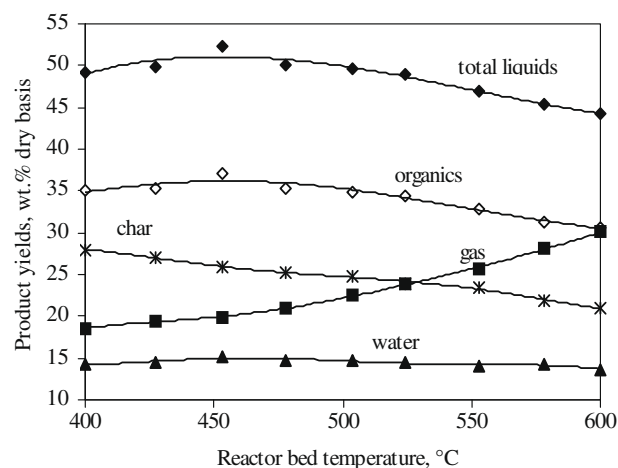
Vapour residence time was investigated first and done with a very narrow feedstock size range due to concerns over feeding issues. It was later found that a wider size range could be fed without causing blockage of the feeding equipment. The experiments, taking temperature into account were therefore done with a slightly wider particle size range.

3. Results and discussion

3.1. Impact of reactor temperature

The experimentally obtained product yields are shown in Figs. 2 and 3. The maximum liquid and organics yields are obtained at 450 °C. As is typical of biomass feedstocks, gas yields increase at higher temperatures, while char yields decrease [6].

The yield trends for carbon monoxide and carbon dioxide are similar to those reported by Scott and Piskorz [7] for aspen-poplar,

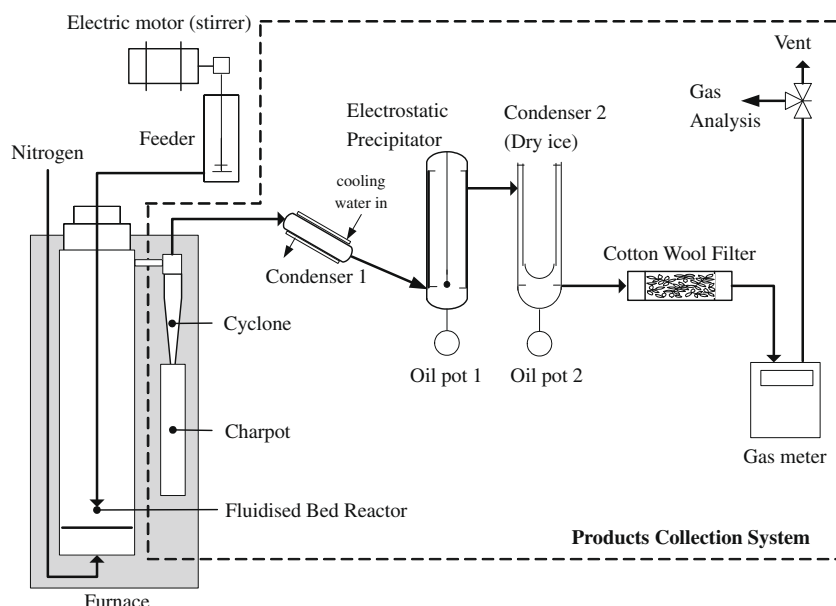
**Fig. 2.** Impact of temperature on pyrolysis yields.

a woody feedstock. They also showed carbon dioxide yields rising less quickly over 550 °C and a sharp rise in carbon monoxide yields at higher temperatures. Their data extended to 700 °C and indicated that at temperatures above 600 °C, carbon monoxide yields became higher than carbon dioxide yields.

3.2. Impact of varying residence time by changing the nitrogen flow rate

The yields resulting from varying the fluidising flow rate are presented in Table 2. It is noted that the fluidising flow rate is given as liters per minute measured at the prevailing room temperature. The hot gas residence time is calculated from the volumetric gas flow rate at operating temperature and pressure and the empty hot space volume, that is the volume of the reactor, cyclone, char pot and pipe work up to the first condenser less the volume of the sand contained in the reactor. Consequently, halving the flow rate does not exactly double the residence time.

It is apparent that there is a maximum liquid yield and a minimum char yield at a residence time of 1.03 s. This is in agreement with similar results obtained by Islam et al. [8] for palm shell. For

**Fig. 1.** Diagram of the fluidised bed pyrolysis system.

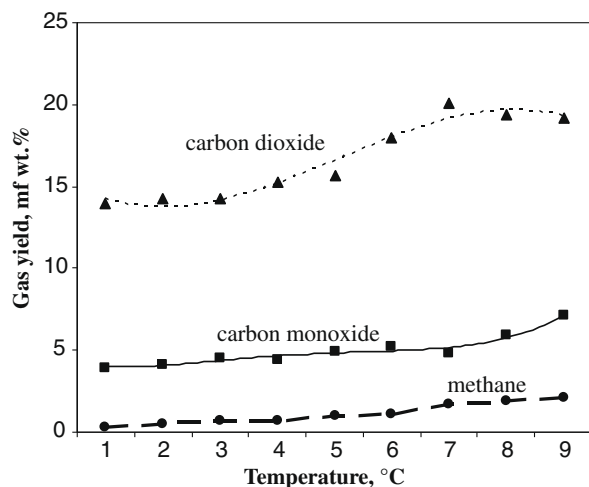


Fig. 3. Impact of temperature on individual gas yields.

Table 2

Yield impact of varying fluidising flow.

Run. no	Vapour residence time (s)	Fluidisation gas flow rate (l/min)	Product yields (mf wt.%)		
			Liquid	Char	Gases
1	0.79	7.0	50.6	27.2	17.9
2	0.96	6.0	51.5	26.5	17.7
3	1.03	5.0	55.1	23.9	18.6
4	1.16	4.5	50.2	25.9	19.1
5	1.23	4.0	47.8	27.5	22.4
6	1.32	3.5	45.3	27.6	25.1

very high fluidising rates it is likely that some of the biomass is blown out of the reactor before it can fully pyrolyse. While the char pot is at the same temperature as the reactor, any vapours released will have extensive contact with char, which is known to increase the char yields [9]. In addition, any tars sticking to the char would contribute to the measured char yield. For longer residence times, secondary reactions will result in the decomposition of the volatile vapours [10]. This clearly shows the sensitivity of yields to variations in fluidising flow and the need to optimise this variable. As this work was performed first, all other experiments were carried out at a constant fluidising flow.

3.3. Combined impact of ash content and particle size

The results obtained are plotted and compared with literature data [6,10–12] for a variety of biomass feedstocks as shown in Fig. 4. The low organic yields found in this work compared well with other high ash content ligno-cellulosic materials, particularly considering the high potassium content of EFB.

Due to feeding difficulties, two of the experiments, for particles sized at <150 μm and 150–250 μm , only had a closure of 90%, while all other experiments had closures above 95%. The low organic yield obtained the highest ash content of particle size range <150 μm , is nevertheless not believed to be purely an artefact of the low closure. All other yields were higher than those obtained for the size range of 300–355 μm , while the organic yield was more than ten percentage lower.

It is also noted that ash content is not the only variable of importance that is impacted by the particle size ranges. Particle size itself may also have an effect on pyrolysis yields, especially for small particles, however, generally it is expected to be minor

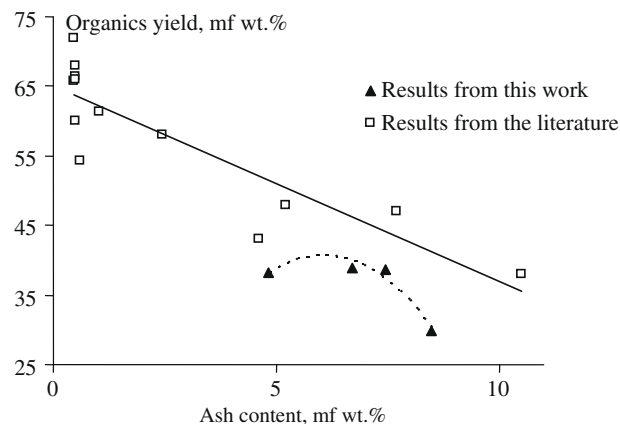


Fig. 4. Comparison of data for EFB with other biomass feedstocks.

[13,6]. It is also possible that the cellulose to lignin ratio of the size fractions differs.

3.4. Physical and chemical properties of products and its potential uses

The pyrolysis liquids produced separated into two phases, a phase predominated by tarry organic compounds and an aqueous phase. The relative shares of the total liquid product yield of the two phases are approximately 60% for the former and 40% for the latter. A comparison of key properties for the two phases with those of wood derived bio-oil, light fuel oil and heavy fuel oil is given in Table 3, which also contains the ultimate analysis of the char product. The value for sulphur was not determined, as there is very little sulphur in the EFB itself. Due to its high water content, the higher heating value of the aqueous phase was not measured.

The viscosity of the aqueous phase is close to that of water, while the organic phase hardly flows at all and, at room temperature has such a high viscosity that it could not be measured with the equipment available to the authors of this research.

The empirical formula of the organics in the organic phase is $\text{CH}_{1.51}\text{O}_{0.14}$. By comparison carbohydrates have an empirical formula of CH_2O , phenol of $\text{CHO}_{0.17}$ and longer chain straight alkanes which approach an empirical formula of CH_2 . It is possible that the organic phase contains a small amount of palm oil, as the organics in the organic phase have a significantly higher hydrogen to carbon ratio than is the case for wood derived slow pyrolysis tars [14].

A precise energy balance was difficult to perform, among other reasons, because the heating value of the aqueous phase was not measured directly. The heating value of the gas is very dependent on the extremely low percentages of hydrocarbons and hydrogen in the gas, which is heavily diluted with nitrogen. Nevertheless, using Dulong's formula for the aqueous phase and for char a rough distribution can be estimated, indicating that of the order of 60% of the energy of the original biomass can be found in the organic phase, 10% in the aqueous phase, 25% in the char and 5% in the gas.

The phase separated liquid product would represent a challenging fuel for boilers and engines, due to the high viscosity of the organic phase and the high water content of the aqueous phase. These could be overcome by upgrading. The addition of polar solvents such as methanol or ethanol represents one of the easiest routes [15] and it was established by the authors of this research that the two phases both readily dissolve in methanol giving a homogeneous single phase product with a low viscosity. Further research would be required to establish the amount of methanol that would have to be added at a minimum to obtain a single phase liquid. The addition of ethanol may also reduce corrosivity and allow removal of the water through low temperature vacuum distil-

Table 3

Characteristics of pyrolysis oil compared to petroleum fuel [15].

	EFB			Wood derived bio-oil	Light fuel oil	Heavy fuel oil
	Organic phase	Aqueous phase	Char			
Elemental analysis (mf wt.%)						
C	69.35	13.83	71.43	32–48	86.0	85.6
H	9.61	11.47	1.8	7–8.5	13.6	10.3
N	0.74	0.14	0.63	<0.4	0.2	0.6
O (by difference)	20.02	74.56	8.72	44–60	0	0.6
S	ND	ND	ND	<0.05	<0.18	2.5
Moisture content (mf wt.%)	7.90	64.01	ND	20–30	0.025	0.1
HHV (MJ/kg)	36.06	ND	ND	–	–	–
LHV (MJ/kg)				13–18	40.3	40.7

ND: not determined.

lation [16], which is otherwise difficult due to the thermal instability of pyrolysis liquids.

Thermochemical upgrading is another possibility, either via gasification and Fischer–Tropsch, where it may be advantageous to gasify bio-oil char slurry rather than the biomass itself [17], or via catalytic steam reforming of the aqueous phase to obtain the hydrogen for hydrogenation of the organic phase [18].

The organic phase may also be used directly in engines, turbines and boilers, if it is first pre-heated to reduce its viscosity, though its thermal instability might limit the temperature it can be brought up to. The aqueous phase could potentially be co-fired to reduce thermal NO_x and enable efficient burning of its dissolved dry matter. As mentioned in the introduction, it might also be possible to obtain useful chemicals, such as phenolic compounds for resin making, as a commercial by-product.

Not all of the char and gas would be required for internal process heat. Some might be used to dry the very wet fresh EFB. Burning the char, which contains nearly all the minerals, would allow recovery of the ash, which is potentially useful as a fertiliser. The gas is very high in carbon dioxide and has a higher heating value of only approximately 4–6 MJ/kg. It might potentially be utilised for carbon sequestration enhanced oil recovery schemes after combustion in oxygen or an oxygen enriched atmosphere [19]. The economics of this would depend on the available carbon price and may not be appealing in the near term, when even near pure carbon dioxide streams available from ethanol or ammonia manufacture are largely vented.

As the fresh EFB is very wet, washing of the EFB prior to fast pyrolysis represents another interesting avenue to enhancing the value of the fuel products obtained, as drying is required already and would therefore add little extra expense [5]. In the thesis of one of the co-authors [20], it is shown that simple soaking in water at room temperature can remove most of the ash, with small biomass losses, giving a feedstock with similar yield characteristics as low ash woody biomass. It is, however, worthwhile to point out that the decline in organic mass yield due to the presence of ash is much greater than the decline in organic energy yield, which is limited. If pyrolysis is primarily used to improve the energy density of biomass for transportation and oil char slurries are acceptable for the end user, the benefit of ash reduction by washing may not be that large, as the combined energy yield of char and oil is essentially unaffected.

4. Conclusion

Oil palm empty fruit bunches have been pyrolysed in a bench scale fluidised bed system. In comparison to the existing literature, this short communication confirms the shape of the yield curves for EFB by observing char, gas, reaction water and organics with a relatively large number of experiments done with temperature

increments of 25 °C each. The data on residence time and particle size reinforced the notion that great care need to be taken when comparing data from different authors obtained with similar but subtly different systems. For example, whether a char pot is heated or not can have a material impact.

Acknowledgments

The authors would like to thank Universiti Sains Malaysia for fully funding the work described in this publication. Most of the experimental work was performed at the University of Aston in Birmingham, while Nurhayati Abdullah was simultaneously a PhD student of the University of Aston in Birmingham and a full time employee of USM.

References

- [1] Foreign Agricultural Service, USDA, October 2005, Oil Seeds Circular, Table 9.
- [2] Foreign Agricultural Service, USDA, November 2005, Oil Seeds Circular.
- [3] Husain Z, Zainal ZA, Abdullah MZ. Analysis of biomass-residue-based cogeneration system in palm oil mills. *Biomass Bioenerg* 2003;24:117–24.
- [4] Bridgwater AV, Peacocke GVC. Fast pyrolysis processes for biomass. *Sustain Renew Energy Rev* 2000;4(1):1–73.
- [5] Abdullah N, Gerhauser H. Bio-oil derived from empty fruit bunches. *Fuel* 2008;87:2606–13.
- [6] Hague RA. The pre-treatment and pyrolysis of biomass for the production of liquids for fuels and speciality chemicals. PhD thesis, Aston University, Birmingham, UK, 1998.
- [7] Scott DS, Piskorz J. The flash pyrolysis of aspen-poplar wood. *Can J Chem Eng* 1982;60:666–74.
- [8] Islam FN, Zailani R, Ani FN. Pyrolytic oil from fluidised bed pyrolysis of oil palm shell and its characterisation. *Renew Energy* 1999;17(1):73–84.
- [9] Várhegyi G, Antal MJ, Jakab E, Szabó P. Kinetic modeling of biomass pyrolysis. *J Anal Appl Pyrolysis* 1997;42(1):73–87.
- [10] Scott DS, Piskorz J. The continuous flash pyrolysis of biomass. *Can J Chem Eng* 1984;62:404–12.
- [11] Scott DS, Piskorz J, Radlein D. Liquid products from the fast pyrolysis of wood and cellulose. *Ind Eng Chem Proc Des Dev* 1985;24:581–8.
- [12] Robinson N Design. Modelling and construction of a Novel ablative fast pyrolysis reactor and product collection system. PhD thesis, Aston University, Birmingham, UK, 2005.
- [13] Di Blasi C. Modelling the fast pyrolysis of cellulosic particles in fluid-bed reactors. *Chem Eng Sci* 2000;55(24):5999–6013.
- [14] EU Contract NNE5-2001-00744, BIOTOX, Final Publishable Report.
- [15] Chiaramonti D, Oasmaa A, Solantausta Y. Power generation using fast pyrolysis liquids from biomass. *Renewable Sustainable Energy Rev* 2007;11(6):1056–86.
- [16] Canadian Patent CA 2165858.
- [17] Zwart RWR, Boerrigter H, van der Drift A. The impact of biomass pretreatment on the feasibility of overseas biomass conversion to fischer–tropsch products. *Energy Fuels* 2006;20:2192–7.
- [18] Wang D, Czernik S, Chornet E. Production of hydrogen from biomass by catalytic steam reforming of fast pyrolysis oils. *Energy Fuels* 1998;12(1):19–24.
- [19] Herzog HJ. What future for carbon capture and sequestration? *Environ Sci Technol* 2001;35(7):148–53.
- [20] Abdullah N. An assessment of pyrolysis for processing empty fruit bunches. PhD thesis, Aston University, Birmingham, UK, 2005.